

# Early Identification of Heart Attack vs Gastritis Using AI/ML

Project ID : R26-IT-012 | Year : 2026



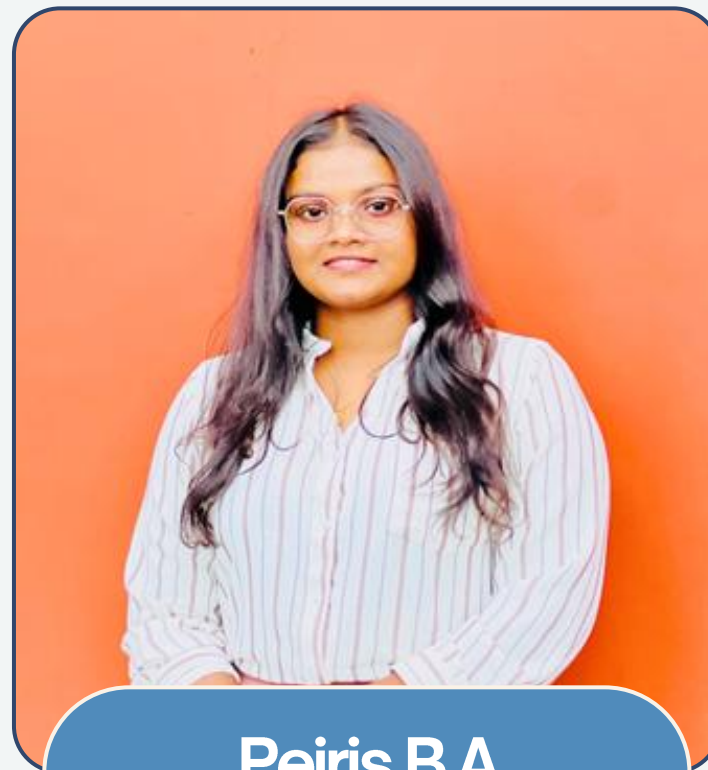
1/09/2026

# Our Team



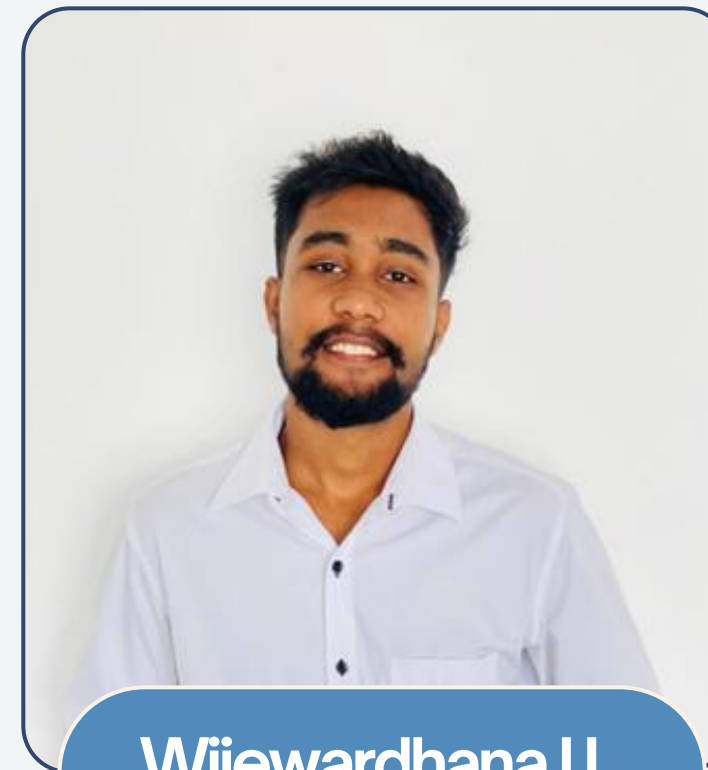
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Heart attacks often resemble gastric conditions, making

# Problem & Motivation

## Clinical Context

- Cardiac and gastric symptoms often overlap.
- Delayed cardiac diagnosis can be life-threatening.
- Diagnosis may require multiple tests and expert review.

## Observed Problems

- Most AI tools focus on a single condition or modality.
- ECG, Echo, angiogram and endoscopy analysis is fragmented.
- Limited integrated support for students and junior clinicians.



# Proposed Solution



## Proposed Solution

A multimodal AI decision-support system that combines screening, cardiac analysis, angiogram analysis, and gastric prediction to help differentiate cardiac and gastric chest pain.

## Key Features

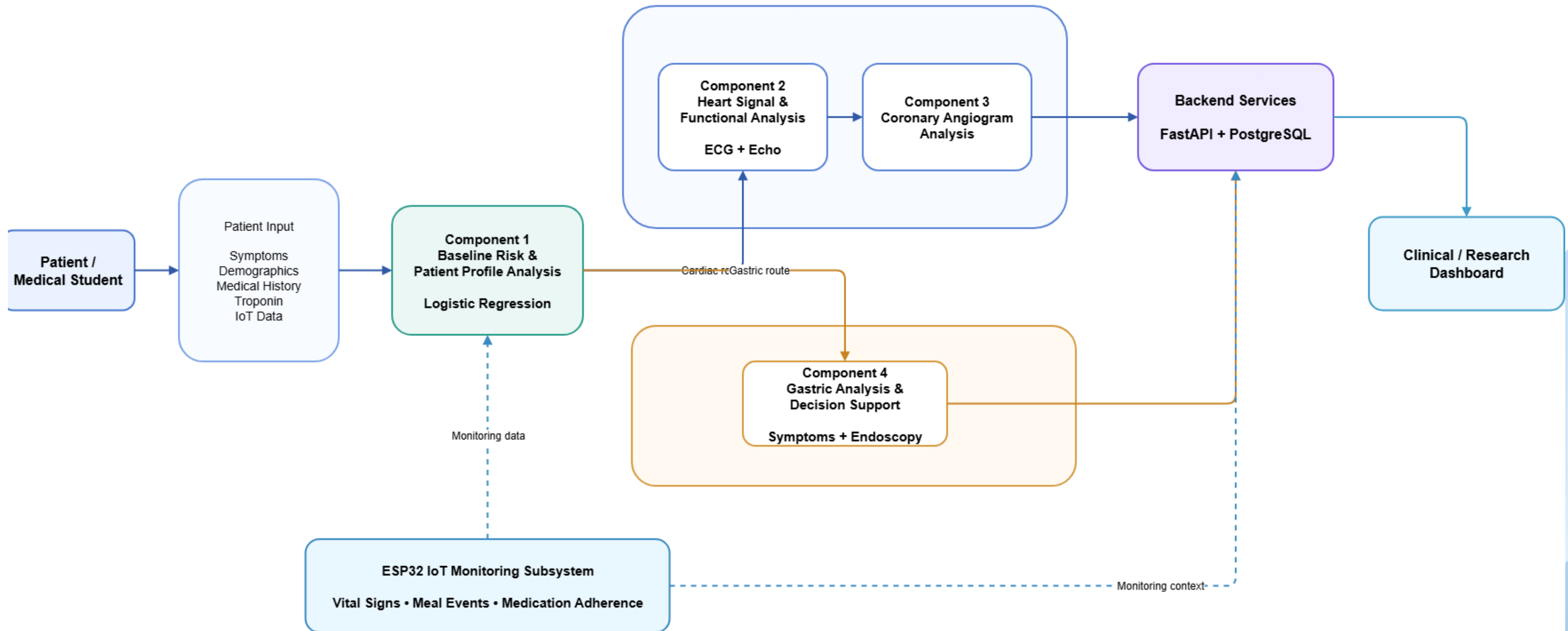
- Cardiac/gastric risk screening
- ECG, Echo & angiogram AI analysis
- Endoscopy-based gastric detection with XAI
- Treatment support & IoT monitoring
- Interpretable AI predictions
- Secure patient data handling
- Fast, scalable architecture
- User-friendly clinical dashboard

# Overall Research Gap

| Capability                               | Typical existing work  | Proposed System                           |
|------------------------------------------|------------------------|-------------------------------------------|
| Cardiac + Gastric differential Diagnosis | Limited / isolated     | Integrated                                |
| Multiple diagnostic modalities           | Mostly single modality | Screening + ECG/Echo + Angio + Exdoscopy  |
| Explainable educational support          | Limited                | Integrated explanations / guided workflow |
| End to end staged decision support       | Rare                   | Yes                                       |



# System Architecture & Integration



# System Architecture & Integration

## Components

1. Baseline Risk & Patient Profile Analysis — IT22193490
2. Heart Signal & Functional Analysis — IT22609380
3. Coronary Angiogram Analysis — IT22116024
4. Gastric Analysis and Decision Support Engine — IT22222640

## Integration Points

- FastAPI REST APIs
- PostgreSQL data storage
- Medical image/video uploads
- Shared dashboard & IoT routing

# Model & Evaluation Evidence

## C1: Baseline Risk

### Logistic Regression

- 1,000 labelled cases
- 59 attributes
- Accuracy 80.5%
- ROC-AUC 0.892

## C2: ECG + ECO

### ResNet + DeepLabV3

- ResNet-50 + R(2+1)D-18 + DeepLabV3
- ECG Accuracy: 78.95%
- ECG Sensitivity: 86.00%
- LVEF & LV Structure Analysis

## C3: Angiogram

### YOLO + 2D CNN

- Accuracy 78.95%
- Sensitivity 86.00%
- Specificity 28.57%
- F1-score 87.76%

## C4: Gastric

### EFFICIENTNET-B4

- Accuracy 85%
- Macro F1 65%
- Macro Sensitivity 73%
- Weighted F1 87%

# Implementation, NFRs & Responsible AI

## Implementation & Integration

- Python, PyTorch & OpenCV model pipelines
- FastAPI backend with PostgreSQL
- Shared API outputs for the dashboard
- ESP32 IoT monitoring support

## Non-Functional Goals

- Easy-to-use staged workflow
- Fast AI/API responses
- Reliable input validation
- Modular and maintainable architecture

## Security, Legal & Ethics

- Anonymized / approved clinical data
- Secure medical-data access
- Report bias, imbalance and model limitations
- Keep clinicians in the decision loop

# Commercialization & Adoption

## Target Users/Market

- Medical students & junior clinicians
- Universities / training institutes
- Teaching hospitals

## Value Proposition

- Integrated cardiac-gastric diagnosis
- Multimodal AI support
- Explainable decision support

## Cost & Feasibility

- Open-source AI stack
- API-based deployment
- Cloud scalable

## Commercial Model

- Institutional SaaS
- University / hospital licensing
- Future API integration

## Adoption Plan

- Pilot – Students & supervisors
- Validate – Usability & model performance
- Deploy – Universities / teaching hospitals
- Scale – Cloud & healthcare integration

# Thank You !